Multimodal Alpha Forecasting and Robust Portfolio Optimization

15.095 Machine Learning Under a Modern Optimization Lens

Catherine Ma (cama12@mit.edu) & **Hanna Zhang** (zeyuzh@mit.edu)

November 29, 2025

1 Introduction

Short-horizon equity return prediction remains one of the most challenging problems in quantitative finance. Daily returns are dominated by noise, exhibit low signal-to-noise ratios, heavily influenced by market microstructure dynamics, and reflect nonlinear interactions among systematic factors, macroeconomic conditions, and firm-specific characteristics. Despite this difficulty, even marginal improvements in predictive accuracy can yield economically meaningful gains when used within a risk-controlled portfolio optimization framework.

The goal of this project is to develop a multimodal forecasting system that integrates complementary sources of information: price-based technical indicators, firm fundamentals, macroeconomic variables, sector and subsector classifications, calendar-time structure, sentiment-derived features and text embeddings from news data. We aim to evaluate which feature groups contribute meaningful predictive power and generate a next-day return forecast suitable for prescriptive portfolio optimization.

2 Data

2.1 Data Sources

Because no single source provides all the information required for forecasting and portfolio construction, we integrate data from multiple independent providers. Our final dataset combines four major categories of information: ddd

- **S&P 500 Constituents:** The list of S&P 500 tickers during our sample period serves as the base universe.
- **Stock Market Data:** Daily OHLCV (Open, High, Low, Close, Volume) data for all S&P 500 stocks from 2016–2023 are obtained from Yahoo Finance.
- **Company Fundamentals:** Firm-level financial metrics (e.g., sales growth, margins, ROA, ROE, leverage, liquidity ratios, valuation ratios) are sourced from WRDS/Compustat.
- **Macroeconomic Indicators:** Broader economic conditions are captured using FRED variables such as GDP growth, unemployment rates, and Treasury yields.
- **News Data:** Daily financial news headlines and summaries are collected from the FNSPID dataset.[1]

2.2 Data Processing Pipeline

After gathering raw inputs, we construct a unified panel dataset at the (ticker, date) level. The main steps are:

1. **Cleaning and Alignment:** All data sources are standardized to daily frequency. We merge by ticker and date, forward-fill fundamentals to match market trading days.
2. **Return Construction:** Using daily closing prices, we compute the next-day stock return, which serves as the prediction target.

[1] `https://huggingface.co/datasets/Zihan1004/FNSPID`

3. **Feature Engineering:** We generate:

   - ***Price-Based Features and Technical Indicators.*** Daily OHLCV data serve as the foundation for constructing short-horizon technical indicators that are commonly used in quantitative trading:
     - Moving averages at several horizons (2-day, 3-day, 1-week, 2-week, 1-month, 1.5-month, 2-month)
     - Exponential moving averages (EMA) with windows of 10, 20, and 50 days
     - Rolling-window return features (e.g., 3-day, 7-day, and 21-day average returns) that capture short-term momentum and drift in recent price movements
     - Volume-derived proxies, such as log volume

     These variables constitute the price-signal feature group, which forms the baseline for many of the models evaluated.
   - ***Fundamental Variables.*** Forward-filled quarterly financial statement items to incorporate firm-level characteristics:
     - Profitability metrics, such as return on assets (ROA), return on equity (ROE), and gross, operating, and net margins
     - Growth metrics, including sales growth, asset growth, equity growth
     - Valuation ratios, such as book-to-market, earnings yield, cash-flow yield, and sales yield
     - Liquidity and leverage ratios, including current ratio, leverage, and cash holdings

     These fundamentals capture cross-sectional variation that evolves more slowly but is often predictive of medium-term equity behavior.
   - ***Macroeconomic Indicators.*** Daily or monthly macro variables aligned to trading days:
     - Market volatility via VIX
     - Yield curve spreads (10Y–2Y, 10Y–3M)
     - Federal Funds Rate
     - Economic activity indicators, including industrial production, unemployment, and GDP

     These variables embed systematic risk dynamics and macro-sensitive return drivers.
   - ***Sector and Subsector Structure.*** Firm-level structure is incorporated using GICS (Global Industry Classification Standard) classifications:
     - One-hot sector indicators
     - One-hot subsector indicators

     These features encode economically meaningful groupings that help models recognize sector-specific return patterns and reduce cross-sectional confounding.

4. **Train–Validation–Test Split:** To avoid look-ahead bias, we split the dataset chronologically:

   - 2016–2021 for training,
   - 2022 for validation,
   - 2023 for out-of-sample testing.

This pipeline produces a complete, aligned dataset suitable for both prediction modeling and portfolio backtesting.

3 Methodology

3.1 Multimodality

Text Features

To incorporate unstructured information alongside traditional financial signals, we extract two types of text-based features from daily news headlines and summaries using the pretrained FinBERT model:

- **Sentiment Scores.** A FinBERT sentiment classifier assigns each news item a positive, negative, or neutral label. For each stock and each day, we aggregate these predictions into summary measures such as mean sentiment, extremal sentiment, and news count.
- **Text Embeddings.** Using the base FinBERT model, we obtain 768-dimensional contextual embeddings for each headline and summary. We compute the daily embedding for each stock by averaging all embeddings associated with its news on that date.

These features capture narrative tone, linguistic signals, and information flow not present in standard price or fundamental data.

Dimension Reduction

Because daily embeddings are high-dimensional, we apply PCA to project them onto 64 components. This retains the majority of the variance while keeping the modeling pipeline computationally manageable for a large cross-section of stocks.

3.2 Stock Next Day Return Prediction

Evaluation Metric

Since the goal of this project is to forecast next-day equity returns in a noisy, low-signal environment, the evaluation framework must reflect the decision-making requirements of a trading system. Although all models generate real-valued return predictions, the primary operational question is whether the model correctly anticipates the *direction* of the next day's movement, as this determines the trading position taken.
To align with this objective, we use **Directional Accuracy (DA)** as the central evaluation metric. DA measures the fraction of predictions whose sign matches the sign of the realized next-day return:

$$\mathrm{DA} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left\{ \operatorname{sign}(\hat{r}_i) = \operatorname{sign}(r_i) \right\},$$

where $\hat{r}_i$ is the predicted next-day return, r_i is the realized next-day return, and $\mathbf{1}\{\cdot\}$ is the indicator function. A DA of 50% corresponds to random guessing in a balanced up/down environment, so improvements of even 1–2 percentage points are economically meaningful in this setting.

3.2.1 Baseline Models

Before introducing flexible machine learning methods, we construct a suite of simple, deterministic forecasting rules. These baselines serve two purposes: (i) to provide interpretable benchmarks grounded in standard quantitative finance heuristics (e.g., momentum and seasonality), and (ii) to quantify how much apparent predictability can arise from rule selection rather than genuine signal.

All baseline models are evaluated on the same S&P 500 daily return panel using the 2016–2022 train–validation period and the 2023 test set.

Single-Rule Baselines

These models apply the same fixed rule across all stocks and dates:

- **Naive persistence:** predicts tomorrow's return as today's return.
- **Moving averages (MA):** simple averages over windows such as 2–60 days.
- **Exponential moving averages (EMA):** exponentially weighted versions of MA rules.
- **Seasonal rules (S1–S7):** include same-day-last-year, same-month-last-year, month-of-year means, day-of-week means, and rolling long-horizon averages (21-day, 63-day).

Directional accuracy for these rules generally clusters around 50%. The persistence rule performs at chance, short-horizon MA/EMA rules yield modest improvements (50.2–50.5% in-sample, reverting toward 50% on test), and several seasonal rules stand out. The day-of-week mean achieves roughly 50.6% train–validation accuracy and 50.9% on the test set, and the month-of-year mean reaches nearly 52% on test, suggesting that coarse calendar structure generalizes better than aggressively tuned short-term momentum rules.

Hybrid Baselines

To relax the "one-rule-for-all" constraint, we evaluate hybrid baselines that allow different segments of the data to select their own preferred single-rule predictor based on 2016–2022 performance:

- **Per-ticker:** each stock selects its best single-rule baseline.
- **Per-month:** each calendar month selects its best rule.
- **Per-day-of-week:** each weekday selects its best rule.

Hybrid Method	**Train+Val DA**	**Test DA**
Per-ticker	0.5185	0.5044
Per-month	0.5144	0.5110
Per-day-of-week	0.5119	0.5040

The per-ticker method attains the highest in-sample accuracy but does not generalize well, while the month-of-year hybrid yields the most stable out-of-sample performance. Notably, the hybrid selections reveal recurring patterns: the day-of-week average is chosen for 143 tickers, long-horizon rolling means for 59, and several MA/EMA windows (50-day EMA, 1-month, 3-day, 14-day) are frequently selected. These patterns indicate that calendar structure and slower-moving return components are the most consistently useful simple predictors, and they inform which feature families are emphasized in the subsequent machine learning models.

Summary

Overall, the baseline results confirm that next-day return prediction is extremely noisy: simple rules rarely exceed 51% accuracy. Nonetheless, seasonality and long-horizon momentum effects provide weak but persistent signals that serve as meaningful benchmarks for evaluating more flexible multimodal models.

3.2.2 XGBoost

3.2.3 CatBoost

3.2.4 LightGBM

3.3 Robust Portfolio Optimization

While predictive accuracy indicates that the model captures return-relevant patterns, turning point forecasts into actionable trades requires an allocation rule that respects risk, diversification, and transaction costs. Simple heuristics such as equal-weighting or top-K selection ignore downside exposures and often produce unstable, overly concentrated portfolios whose performance exaggerates the value of the underlying signal.

To obtain realistic and risk-aware allocations, we employ a robust Conditional Value-at-Risk (CVaR) optimization layer. CVaR determines the weight assigned to each equity by evaluating how the portfolio would behave under a rolling set of recent historical return scenarios. For each day's prediction vector, the optimizer searches for weights that increase exposure to stocks with high predicted returns, but penalize those that contribute heavily to losses in the worst tail of the return distribution. In practice, this yields portfolios that favor assets with both attractive forecasts and stable downside behavior, while enforcing constraints such as long-only positions, position-size caps, and turnover limits. The result is a set of portfolio weights that are smoother, more diversified, and more robust than those produced by naïve allocation rules.

3.3.1 CVaR Formulation

Rather than relying on variance as a proxy for risk, we use CVaR to focus directly on large downside losses. CVaR measures the expected loss in the worst $(1-\alpha)$ fraction of outcomes and remains well-behaved even when returns are skewed or heavy-tailed. Given portfolio weights w and a set of recent scenario returns $\{r_t\}_{t=1}^{T}$, CVaR at confidence level $\alpha = 0.95$ is written using auxiliary variables v (the VaR cutoff) and z_t (tail losses):

$$\mathrm{CVaR}_\alpha(w) = v + \frac{1}{(1-\alpha)T}\sum_{t=1}^{T} z_t, \qquad z_t \geq 0, \quad z_t \geq -(r_t^\top w) - v.$$

This linear representation makes the CVaR objective directly optimizable.

3.3.2 Rolling Scenario Construction

The scenario matrix is built from the most recent 60 days of realized cross-sectional returns. Using a rolling window allows the optimizer to reflect current market conditions—including changes in volatility, correlations, and downside risk—without assuming a particular return distribution. As market regimes shift, the CVaR optimizer naturally updates its estimate of tail risk and adjusts portfolio weights accordingly.

3.3.3 Optimization Objective and Constraints

At each date t, the optimizer selects weights for each equity by solving

$$\max_{w} \quad \hat{\mu}_t^\top w \;-\; \lambda\,\mathrm{CVaR}_{0.95}(w) \;-\; \mathrm{TC}\cdot\|w - w_{t-1}\|_1,$$

where $\hat{\mu}_t$ are the model's predicted next-day returns (scaled for numerical stability), λ controls aversion to tail risk, and the ℓ_1 turnover term penalizes excessive rebalancing and incorporates proportional transaction costs.

We impose the following practical trading constraints:

- **Long-only and fully invested:** $w_i \geq 0$, $\sum_i w_i = 1$.
- **Position limit:** $w_i \leq 2\%$ to prevent concentration.
- **Turnover limit:** $\|w - w_{t-1}\|_1 \leq 10\%$ to avoid hyperactive trading.
- **Transaction cost penalty:** proportional cost of 5 bps per unit turnover.

These constraints make the optimization problem both realistic and stable, preventing extreme weight allocations that often arise in unconstrained ML portfolios.

3.3.4 Backtesting Procedure

For each model's cross-sectional predictions, we build a $T \times N$ matrix of predicted returns and a matching matrix of realized one-day-ahead returns. The portfolio is rebalanced daily using the CVaR optimizer. Performance is reported using standard metrics:

$$\text{Sharpe ratio,} \quad \text{annualized return,} \quad \text{annualized volatility,} \quad \text{and maximum drawdown.}$$

Importantly, CVaR-based optimization acts as a "stress test" for the machine learning signals: only predictions that are stable, consistent across the cross-section, and resistant to noise will survive the downside-risk penalty and turnover constraints.

4 Results

To compare the predictive contribution of different input modalities, we evaluate five LightGBM models trained on progressively richer feature sets:

- **Fundamentals:** Day-Month-Sector (DMS), log market capitalization ($\log\_mktcap$), book-to-market ratio (bm), return on assets (roa_ttm), and return on equity (roe_ttm).
- **Mean Sentiment:** DMS, and mean sentiment score from news data.
- **Mean Sentiment + Embeddings:** DMS, mean sentiment score, and text embeddings.
- **News Momentum:** DMS and a full set of leak-free sentiment momentum predictors. For each raw sentiment variable $f \in \{\text{mean}, \text{max}, \text{min}, \text{sum}, \text{count}\}$, the feature set includes: f, f_{lag1}, f_{roll3}, f_{roll5}, f_{vol5}, f_{shock}, and $f_{\text{surprise3}}$.
- **News Momentum + Embeddings:** DMS, News Momentum, and text embeddings.

Directional Accuracy for LightGBM Models with Different Feature Sets

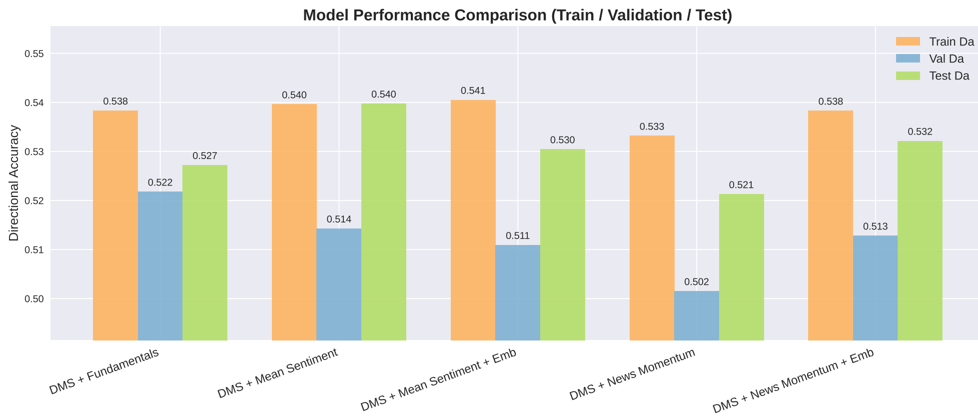

Figure 1: Directional accuracy for LightGBM models trained with different feature sets.

Interpretation. Across all models, directional accuracy remains in the narrow range of 0.51–0.54, consistent with well-known limits on daily return predictability. Several patterns emerge. First, all models outperform a naïve random-guessing benchmark (50%), confirming that even simple tabular signals contain economically meaningful information. Second, the *Fundamentals* and *Mean Sentiment* models achieve the highest test accuracy (approximately 0.540), indicating that slow-moving valuation ratios and high-level news sentiment indicators are the most stable predictors in our dataset. Third, adding PCA-based text embeddings does not improve performance and, in some cases, slightly reduces accuracy; this suggests that embeddings introduce additional noise relative to their marginal predictive value in daily horizons. Finally, the full *News Momentum* feature set yields modest gains over mean sentiment alone, but the improvement is not consistently reflected on the validation and test splits, which indicates overfitting.

Portfolio Performance under CVaR Optimization with Different Feature Sets

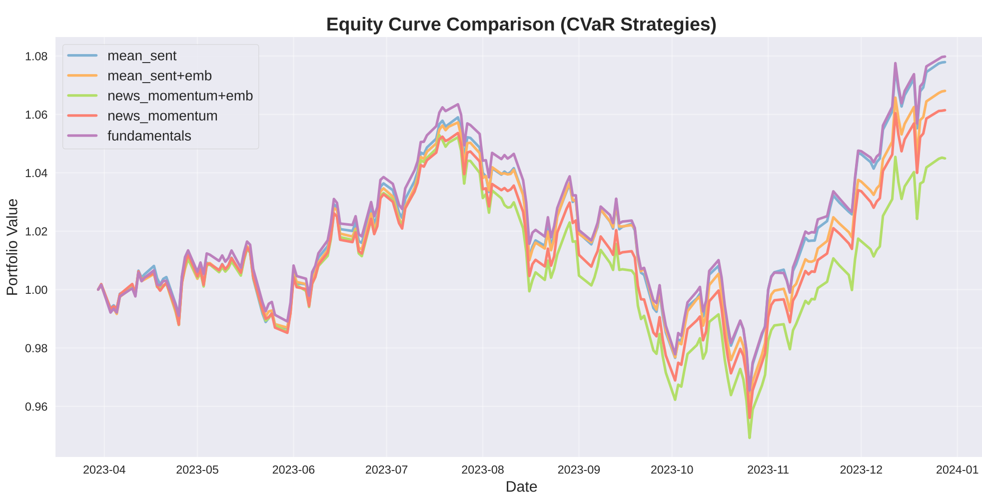

Figure 2: Equity curves for portfolios using CVaR optimization with different feature sets.

Table 1: Performance and Information Coefficient (IC) Statistics for Different Feature Sets

Model	**Sharpe**	**Ann. Return (%)**	**Mean IC**	**IR**
DMS + Fundamentals	**1.166**	**10.84**	**0.02834**	**0.1669**
DMS + Mean Sentiment	**1.142**	**10.58**	**0.02685**	**0.1691**
DMS + Mean Sentiment + Embeddings	0.989	9.23	0.02067	0.1629
DMS + News Momentum	0.895	8.32	0.02786	0.1906
DMS + News Momentum + Embeddings	0.654	6.07	0.01457	0.1312

Interpretation. Table 1 summarizes both economic performance and statistical signal quality. The *Sharpe ratio* and *annualized return* reflect how well each model performs when translated into a trading strategy, while the *Information Coefficient (IC)* measures the cross-sectional correlation between predicted and realized returns. The *Information Ratio (IR)* indicates the stability of this signal over time.

Fundamentals and mean sentiment deliver the strongest performance, with the highest Sharpe ratios (≈ 1.14–1.17) and mean IC values (≈ 0.027–0.028). This suggests that slow-moving fundamentals and aggregate sentiment remain the most reliable predictors at a daily horizon. Adding text embeddings consistently reduces both Sharpe and IC, indicating that high-dimensional embeddings introduce noise rather than improving predictability. News momentum features offer moderate improvements in IC stability (higher IR) but do not outperform fundamentals in realized portfolio performance. Overall, simpler feature sets provide clearer and more stable signals than embedding-augmented models.

5 Discussion and Conclusion

6 Contribution